



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2024-25
COURSE HANDOUT

Date: 02.08.2024

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : CE F211
Course Title : **Mechanics of Solids**
Class Room Nos : 1201 (L), 1203 (T1), 1226 (T2)

Instructor-in-Charge : DIPENDU BHUNIA

Tutorial/Practical Instructors : Potdar Namdev Mukund

1. Course Description:

Fundamental principles of mechanics; Introduction to the mechanics of deformable bodies; Forces and Moments transmitted by slender members; Stress- Strain; Stress-Strain Temperature relations; Torsion; stresses and deflections due to bending; Failure theories, Buckling of columns using Euler's Theory.

2. Scope and Objective of the Course:

The subject of mechanics of solids deals with determination of strength, deformation and stability of structural and machine elements. The methods are based on Laws of Newtonian mechanics, applied to bodies in static equilibrium, geometry and experimentation. These laws are applied to simple situations with engineering judgment to arrive at results of significance to the designer.

At the end of the course the student will be in a position to design and analyze simple structural elements, which involve calculation of stress, strain and deformation. This is an essential feature in any design process.

3. Text Books:

1. S. H. Crandall et al., An Introduction to Mechanics of Solids, McGraw-Hill International editions.

4. Reference Books:

1. Mechanics of Materials; F. P. Beer, E. R. Johnston, J. T. DeWolf and D. F. Mazurek, Fifth Edition, 2011, McGraw-Hill Education Pvt. Ltd., India.
2. Engineering Mechanics of Solids, Egor P. Popov, 2nd Edition, 1998, PHI, India.
3. Mechanics of Materials, James M. Gere, 6th Edition, 2006, Thomson Books/Cole, USA.

5. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
1-3	Introduction, principles of mechanics, concept of force & moment, equilibrium conditions, concept of two & three force members, free body diagram, friction	Ch. 1 (TB)	Fundamental principles of mechanics
4-5	Numerical problems	Ch.1 (TB)	Fundamental principles of mechanics



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
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6-8	Analysis of deformable bodies, uniaxial loading & deformation, statically determinate truss, hoop stresses in thin cylindrical shells, Castigliano's theorem.	Ch. 2 (TB)	Introduction to mechanics of deformable bodies
9-10	Numerical problems	Ch.2 (TB)	Introduction to mechanics of deformable bodies
11-12	Introduction forces & moments acting on a section of a member, distributed loads & resultant of distributed loads	Ch. 3 (TB)	Forces & moments transmitted by slender members
13-14	Differential equilibrium approach, Singularity functions	Ch. 3 (TB)	Forces & moments transmitted by slender members
15-16	Numerical problems	Ch.3 (TB)	Forces & moments transmitted by slender members
17-18	Introduction, stress, plane stress, equilibrium of an element in plane stress, Mohr circle representation of a plane stress, general state of stress.	Ch. 4 (TB)	Stress & Strain
19-20	Analysis of deformations, strain components, relation between strain & displacement, strain component associated with arbitrary set of axis, Mohr circle representation of plane strain, general state of strain	Ch. 4 (TB)	Stress & Strain
21-22	Numerical problems	Ch.4 (TB)	Stress & Strain
23-24	Introduction, tensile test, idealization of stress strain curve, elastic stress strain relation	Ch. 5 (TB)	Stress-Strain-Temperature relations
25-26	Thermal strain, complete equations of elasticity, strain energy in a elastic body, criteria of initial yielding	Ch. 5 (TB)	Stress-Strain-Temperature relations
27	Numerical problems	Ch.5 (TB)	Stress-Strain-Temperature relations
28-29	Introduction, geometry of deformation of a twisted circular shaft, stress strain relations, equilibrium requirements, stresses & deformations in twisted elastic circular shaft, torsion of elastic hollow circular shaft, combined stresses, strain energy due to torsion, yielding in torsion & Numerical	Ch.6 (TB)	Torsion
30	Numerical problems	Ch.6 (TB)	Torsion
31-32	Introduction, deformation in pure bending, stress-strain relations, equilibrium	Ch. 7	Stresses due to bending



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

	requirements, stresses & deformations in pure bending	(TB)	
34	Stresses due to shear force and bending moment, combined stresses, strain energy due to bending, yielding in bending	Ch. 7 (TB)	Stresses due to bending
35	Numerical problems	Ch. 7	Stresses due to bending
36-37	Introduction, moment-curvature-relations, integration of moment-curvature relations, superposition	Ch. 8 (TB)	Deflections due to bending
38	Load-deflection differential equation, Energy Methods, Numerical problems	Ch. 8 (TB)	Deflections due to bending
39	Numerical problems	Ch. 8	Deflections due to bending
40-41	Introduction, elastic stability, examples of instability, elastic stability of flexible columns	9.1-9.4 (T.B.)	Stability of equilibrium buckling
42	Numerical problems	Ch.9	Stability of equilibrium buckling

6. Evaluation Scheme:

Sl. No.	Evaluation Component	Duration	Weightage	Date, Time & Venue	Nature of Component
1.	Mid-semester	1.5 hrs	30%		CB
2.	Tutorials/ Quiz	—	25%		—
3.	Compre. Exam.	3 hrs	45%		OB

7. Chamber Consultation Hour: Will be announced by instructors individually in the class

8. Notices: Notice, if any, concerning the course will be displayed on the Nalanda LMS.

9. Make-up Policy: Make-up will be granted only to genuine cases. For cases related to illness, proper documentary evidence is essential. Prior permission is necessary if student is out of station on the test date.
No make up for tutorial tests.

Instructor-in-charge
Course No. CE F211